

## SPRINT 3

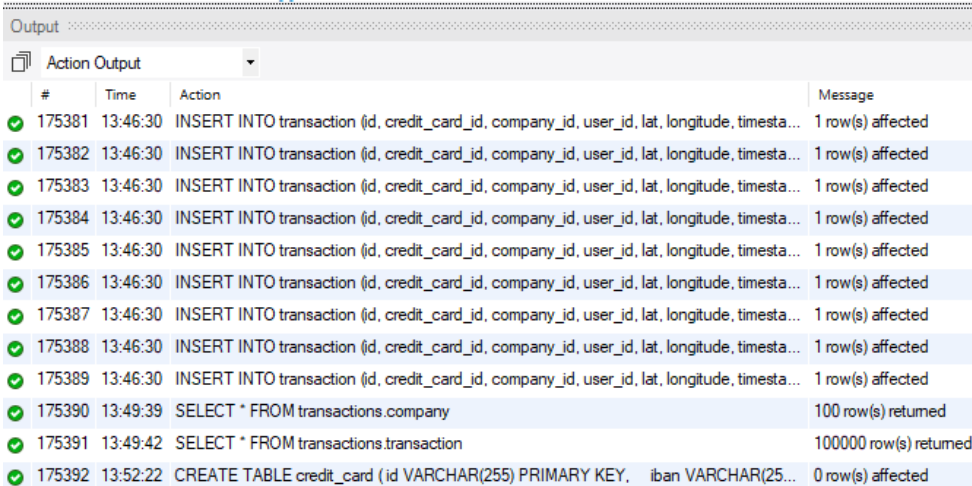
### Nivel 1

#### Ejercicio 1

→ Tu tarea es diseñar y crear una tabla llamada "credit\_card" que almacene detalles cruciales sobre las tarjetas de crédito. La nueva tabla debe ser capaz de identificar de forma única cada tarjeta y establecer una relación adecuada con las otras dos tablas ("transaction" y "company"). Después de crear la tabla, será necesario que ingreses la información del documento denominado "datos\_introducir\_credit". Recuerda mostrar el diagrama y realizar una breve descripción de este.

- Creación de la tabla:

```
5 #Crear tabla credit card:
6 CREATE TABLE credit_card (
7     id VARCHAR(255) PRIMARY KEY,
8     iban VARCHAR(255),
9     pan VARCHAR(255),
10    pin VARCHAR(255),
11    cvv VARCHAR(255),
12    expiring_date VARCHAR(255)
13 );
```



The screenshot shows a database interface with a code editor and an 'Output' window. The code editor contains the SQL statement to create the 'credit\_card' table. The 'Output' window shows a list of database actions, including several 'INSERT INTO transaction' statements and a 'CREATE TABLE credit\_card' statement.

| #      | Time     | Action                                                                                       | Message                |
|--------|----------|----------------------------------------------------------------------------------------------|------------------------|
| 175381 | 13:46:30 | INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timesta... | 1 row(s) affected      |
| 175382 | 13:46:30 | INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timesta... | 1 row(s) affected      |
| 175383 | 13:46:30 | INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timesta... | 1 row(s) affected      |
| 175384 | 13:46:30 | INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timesta... | 1 row(s) affected      |
| 175385 | 13:46:30 | INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timesta... | 1 row(s) affected      |
| 175386 | 13:46:30 | INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timesta... | 1 row(s) affected      |
| 175387 | 13:46:30 | INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timesta... | 1 row(s) affected      |
| 175388 | 13:46:30 | INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timesta... | 1 row(s) affected      |
| 175389 | 13:46:30 | INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timesta... | 1 row(s) affected      |
| 175390 | 13:49:39 | SELECT * FROM transactions.company                                                           | 100 row(s) returned    |
| 175391 | 13:49:42 | SELECT * FROM transactions.transaction                                                       | 100000 row(s) returned |
| 175392 | 13:52:22 | CREATE TABLE credit_card ( id VARCHAR(255) PRIMARY KEY, iban VARCHAR(25...                   | 0 row(s) affected      |

Aquí se crea la tabla vacía como se muestra en el screenshot siguiente:

1 • `SELECT * FROM transactions.credit_card;`

< **Result Grid** Filter Rows:  | Edit: | Export/Import: | Wrap Cell Content:

|   | id   | iban | pan  | pin  | cvv  | expiring_date |
|---|------|------|------|------|------|---------------|
| * | NULL | NULL | NULL | NULL | NULL | NULL          |

credit\_card 1 x

Output

Action Output

| #        | Time     | Action                                                                    | Message           |
|----------|----------|---------------------------------------------------------------------------|-------------------|
| ✓ 175392 | 13:52:22 | CREATE TABLE credit_card (id VARCHAR(255) PRIMARY KEY, iban VARCHAR(25... | 0 row(s) affected |
| ✓ 175393 | 13:53:29 | SELECT * FROM transactions.credit_card                                    | 0 row(s) returned |

- Cargar los datos que nos dan en Moodle:

Simplemente, descargamos el documento “datos\_introducir\_sprint3\_credit.sql” y en sql se corre el script completo para cargar los datos en la tabla credit\_card que sea creó en el paso anterior y obtenemos una tablas con datos como se muestra a continuación:

```
1 • SELECT * FROM transactions.credit_card;
```

| Result Grid        |          |                          |                  |      |     |               |  |
|--------------------|----------|--------------------------|------------------|------|-----|---------------|--|
| Filter Rows:       |          |                          |                  |      |     |               |  |
| Edit:              |          |                          |                  |      |     |               |  |
| Export/Import:     |          |                          |                  |      |     |               |  |
| Wrap Cell Content: |          |                          |                  |      |     |               |  |
|                    | id       | iban                     | pan              | pin  | cvv | expiring_date |  |
| ▶                  | CcS-4857 | XX4857591835292505850771 | 2314242385113924 | 1819 | 467 | 09/27/25      |  |
|                    | CcS-4858 | XX8581768137002436094025 | 6582720299715533 | 3964 | 817 | 12/28/28      |  |
|                    | CcS-4859 | XX7826930491423553609370 | 8861684536289642 | 4983 | 277 | 11/26/26      |  |
|                    | CcS-4860 | XX5559590368835304645299 | 2481155515498459 | 6876 | 661 | 07/27/27      |  |
|                    | CcS-4861 | XX2035182877195191627307 | 1308930301149557 | 5710 | 398 | 04/25/26      |  |
|                    | CcS-4862 | XX4774721462463645409758 | 6715617009807829 | 4042 | 174 | 11/27/26      |  |
|                    | CcS-4863 | XX1476829664245046207111 | 3140879819451394 | 5969 | 449 | 12/27/29      |  |
|                    | CcS-4864 | XX8380298893385731196159 | 5793672133649114 | 8481 | 139 | 02/28/26      |  |
|                    | CcS-4865 | XX7085078596101025280599 | 5101552687251312 | 7847 | 903 | 11/25/28      |  |
|                    | CcS-4866 | XX4792859188206596406839 | 8080768801072613 | 9271 | 961 | 02/28/25      |  |
|                    | CcS-4867 | XX6038298816319374853717 | 7761849537661098 | 4820 | 862 | 11/30/25      |  |
|                    | CcS-4868 | XX3929101617300533068044 | 4160419663151801 | 6860 | 549 | 08/25/25      |  |
|                    | CcS-4869 | XX1566712096111531886465 | 5892930716233310 | 4775 | 448 | 09/29/25      |  |

credit\_card 1 x

Output

Action Output

| #        | Time     | Action                                                                                    | Message              |
|----------|----------|-------------------------------------------------------------------------------------------|----------------------|
| ✓ 180393 | 13:56:27 | INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9581', '... | 1 row(s) affected    |
| ✓ 180394 | 13:58:32 | SELECT * FROM transactions.credit_card                                                    | 5000 row(s) returned |

- Unión de la tabla al modelo por medio de la indicación de la fk

Para crear la relación de la nueva tabla creada con el resto de las tablas, queremos indicar cual va a ser la foreign key de nuestra nueva tabla, cual se va a unir con la primary key de la tabla transaction, para ello se utiliza el siguiente query:

```

43 • ALTER TABLE transaction
44 ADD CONSTRAINT FK_transaction_creditcard
45 FOREIGN KEY (credit_card_id)
46 REFERENCES credit_card(id);
47
48

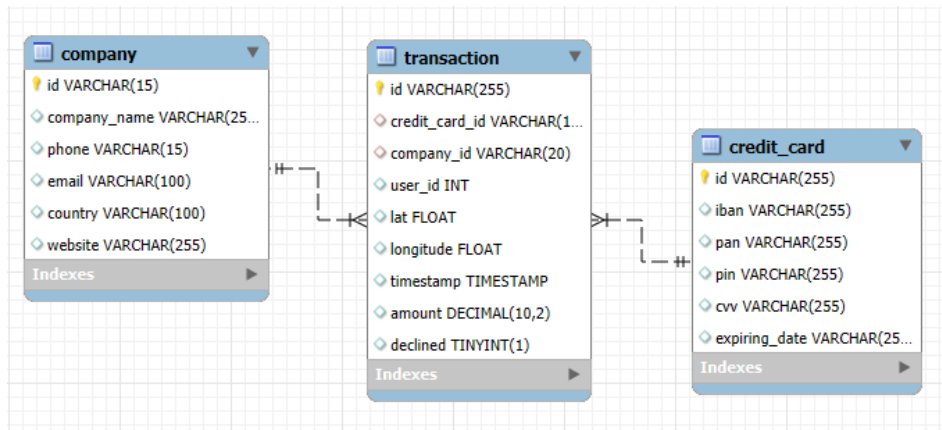
```

Output

Action Output

| #        | Time     | Action                                                                        | Message                                                          |
|----------|----------|-------------------------------------------------------------------------------|------------------------------------------------------------------|
| ✓ 180394 | 13:58:32 | SELECT * FROM transactions.credit_card                                        | 5000 row(s) returned                                             |
| ✓ 180395 | 14:00:01 | ALTER TABLE transaction ADD CONSTRAINT FK_transaction_creditcard FOREIGN K... | 100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0 |

- El diagrama que obtenemos es el siguiente:



#### - Breve descripción:

Al modelo anterior se le ha añadido una nueva tabla (credit card), la cual se ha creado desde cero, se le han agregado los datos proporcionados y posteriormente se ha unido al modelo a través de las PK y FK correspondientes como se ha mostrado en el paso a paso anterior. Este nuevo modelo cuenta con tres tablas, donde la tabla transaction hace las de tabla central, es decir, tanto la tabla company, como la tabla credit card tienen relación 1 a N hacia la tabla transaction. Lo que quiere decir que una compañía puede generar varias transacciones y, por su parte, una tarjeta de crédito puede generar muchas transacciones.

### Ejercicio 2

El departamento de Recursos Humanos ha identificado un error en el número de cuenta asociado a su tarjeta de crédito con ID CcU-2938. La información que debe mostrarse para este registro es: TR323456312213576817699999. Recuerda mostrar que el cambio se realizó.

- con el siguiente query buscamos el registro ID CcU-2938 al que está asociado el error generado:

```

48      #Ejercicio 2
49      #error asociado al numero de cuenta asociado a la tarjeta de credito con id CcU-2938
50      #iban correcto: TR323456312213576817699999
51      • SELECT *
52      FROM credit_card as cc
53      WHERE id ='CcU-2938'

```

| Result Grid                       |          |                            |                  |      |      |               |
|-----------------------------------|----------|----------------------------|------------------|------|------|---------------|
| Filter Rows: <input type="text"/> |          |                            |                  |      |      |               |
| Edit:                             |          |                            |                  |      |      |               |
| Export/Import:                    |          |                            |                  |      |      |               |
| Wrap Cell Content:                |          |                            |                  |      |      |               |
|                                   | id       | iban                       | pan              | pin  | cvv  | expiring_date |
| ▶                                 | CcU-2938 | TR301950312213576817638661 | 5424465566813633 | 3257 | 984  | 10/30/22      |
| *                                 | NULL     | NULL                       | NULL             | NULL | NULL | NULL          |

credit\_card 6 x

Output



Action Output

| #        | Time     | Action                                                                         | Message                |
|----------|----------|--------------------------------------------------------------------------------|------------------------|
| ✓ 185420 | 14:28:37 | ALTER TABLE transaction ADD CONSTRAINT FK_t_cc FOREIGN KEY (credit_card_id)... | 100000 row(s) affected |
| ✓ 185421 | 14:29:10 | SELECT * FROM credit_card as cc WHERE id ='CcU-2938'                           | 1 row(s) returned      |

- La información que debe mostrarse para este registro es: TR323456312213576817699999, por lo que se emplea la primera query para realizar el cambio de campo y en el segundo query revisamos que el cambio se haya realizado correctamente.

```

56 • UPDATE credit_card as cc
57 SET iban = 'TR323456312213576817699999'
58 WHERE id = 'CcU-2938'
59 ;
60 • SELECT *
61 FROM credit_card as cc
62 WHERE id = 'CcU-2938'
63 ;

```

| Result Grid                               |          |                            |                  |      |      |               |
|-------------------------------------------|----------|----------------------------|------------------|------|------|---------------|
| Filter Rows: <input type="text"/>         |          |                            |                  |      |      |               |
| Edit:  Export/Import:  Wrap Cell Content: |          |                            |                  |      |      |               |
|                                           | id       | iban                       | pan              | pin  | cvv  | expiring_date |
| ▶                                         | CcU-2938 | TR323456312213576817699999 | 5424465566813633 | 3257 | 984  | 10/30/22      |
| *                                         | NULL     | NULL                       | NULL             | NULL | NULL | NULL          |

| credit_card 7 × |          |                                                                                  |                                                          |  |
|-----------------|----------|----------------------------------------------------------------------------------|----------------------------------------------------------|--|
| Output          |          |                                                                                  |                                                          |  |
| Action Output   |          |                                                                                  |                                                          |  |
| #               | Time     | Action                                                                           | Message                                                  |  |
| ✓ 185422        | 14:30:51 | UPDATE credit_card as cc SET iban = 'TR323456312213576817699999' WHERE id = '... | 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0 |  |
| ✓ 185423        | 14:30:54 | SELECT * FROM credit_card as cc WHERE id = 'CcU-2938'                            | 1 row(s) returned                                        |  |

### Ejercicio 3

En la tabla "transaction" ingresa una nueva transacción con la información proporcionada en Moodle.

Al intentar añadir la información proporcionada, me encontré con que me arrojaba un error, debido a que el id de la compañía que se estaba agregando en la tabla transaction no existe en la tabla company, por ende y debido a la relación entre las tablas, no se puede insertar una nueva fila en la tabla transaction hasta haber creado ese nuevo campo en la tabla company. Frente a este problema, consideré dos opciones: primero, encontré en la documentación la opción de eliminar (DROP) la relación que existe entre las tablas para así crear la nueva fila de campos en la tabla transaction. Esta opción decidí descartarla pues al eliminar la relación desharía el modelo y podría causar mayores confusiones. Aparte me costó un montón hacer el modelo como para ahora deshacerlo.

La segunda opción, por la cual opté, es añadir la fila en la tabla compañía y dejar la mayoría de los campos vacíos, pues no se cuenta con la información de la compañía más allá de su id. Igualmente, en la información proporcionada hace falta la información para un campo en la nueva fila transaction, timestamp que también opté por dejar en blanco.

- Creación del nuevo id en la tabla company y la revisión de que se añadió correctamente:

```

91 #Añadir nueva fila a la tabla company con el id 'b-9999'
92 • INSERT INTO company (id, company_name, phone, email, country, website)
93 VALUES ('b-9999', NULL, NULL, NULL, NULL, NULL)
94 ;
95 #revisar que se añadió correctamente
96 • SELECT *
97 FROM company
98 WHERE id='b-9999';

```

| Result Grid    |        |              |       |       |         |         |
|----------------|--------|--------------|-------|-------|---------|---------|
| Filter Rows:   |        |              |       |       |         |         |
| Edit:          |        |              |       |       |         |         |
| Export/Import: |        |              |       |       |         |         |
| Wr             |        |              |       |       |         |         |
|                | id     | company_name | phone | email | country | website |
| ▶              | b-9999 | NULL         | NULL  | NULL  | NULL    | NULL    |
| *              | NULL   | NULL         | NULL  | NULL  | NULL    | NULL    |

| company 1 x   |          |                                         |   |
|---------------|----------|-----------------------------------------|---|
| Output        |          |                                         |   |
| Action Output |          |                                         |   |
| #             | Time     | Action                                  |   |
| ✓ 10          | 10:47:48 | select * from company where id='b-9999' | 1 |
| ✓ 11          | 10:48:56 | SELECT * FROM company WHERE id='b-9999' | 1 |

Vale, pues pensaba que al crear esta fila sería suficiente para que no se generara el error, pero nuevamente obtuve el mismo error 1452 (<https://stackoverflow.com/questions/21659691/error-1452-cannot-add-or-update-a-child-row-a-foreign-key-constraint-fails>). Bueno, pues volviendo a revisar el modelo y los campos que se deben ingresar en la nueva fila de la tabla transaction, me di cuenta de que tampoco existe el id 'CcU-9999' en la tabla credit card. Así que repetí el mismo proceso de crear una nueva fila con este id y el resto de los campos vacíos para la tabla credit\_card.

- Creación del nuevo id en la tabla credit card y revision de que se realizó correctamente:

```

99      #Añadir nueva fila a la tabla credit caard con el id CcU-9999
100 •   INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date)
101     VALUES ('CcU-9999', NULL, NULL, NULL, NULL, NULL)
102     ;
103     #revisar que se añadió correctamente
104 •   SELECT *
105     FROM credit_card
106     WHERE id='CcU-9999';

```

Result Grid

|   | id       | iban | pan  | pin  | cvv  | expiring_date |
|---|----------|------|------|------|------|---------------|
| ▶ | CcU-9999 | NULL | NULL | NULL | NULL | NULL          |
| * | NULL     | NULL | NULL | NULL | NULL | NULL          |

credit\_card 3 x

Output

Action Output

| #    | Time     | Action                                          | Message           |
|------|----------|-------------------------------------------------|-------------------|
| ✓ 18 | 11:13:26 | select * from credit_card where id = 'CcU-9999' | 1 row(s) returned |
| ✓ 19 | 11:13:48 | SELECT * FROM credit_card WHERE id='CcU-9999'   | 1 row(s) returned |

- Nuevo ingreso de la tabla transaction como pide la consulta y la revisión de que se generó correctamente:

```

108     #ahora si que puedo añadir la info que me dan en transaction
109 •   INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, timestamp, amount, declined)
110     VALUES ('108B1D1D-5B23-A76C-55EF-C568E49A99DD', 'CcU-9999', 'b-9999', '9999', 829.999, -117.999, NULL, 111.11, 0)
111     ;
112     #revisar que se añadió correctamente
113 •   SELECT*
114     FROM transaction
115     WHERE credit_card_id='CcU-9999';

```

Result Grid

|   | id                                   | credit_card_id | company_id | user_id | lat     | longitude | timestamp | amount | declined |
|---|--------------------------------------|----------------|------------|---------|---------|-----------|-----------|--------|----------|
| ▶ | 108B1D1D-5B23-A76C-55EF-C568E49A99DD | CcU-9999       | b-9999     | 9999    | 829.999 | -117.999  | NULL      | 111.11 | 0        |
| * | NULL                                 | NULL           | NULL       | NULL    | NULL    | NULL      | NULL      | NULL   | NULL     |

transaction 5 x

Output

Action Output

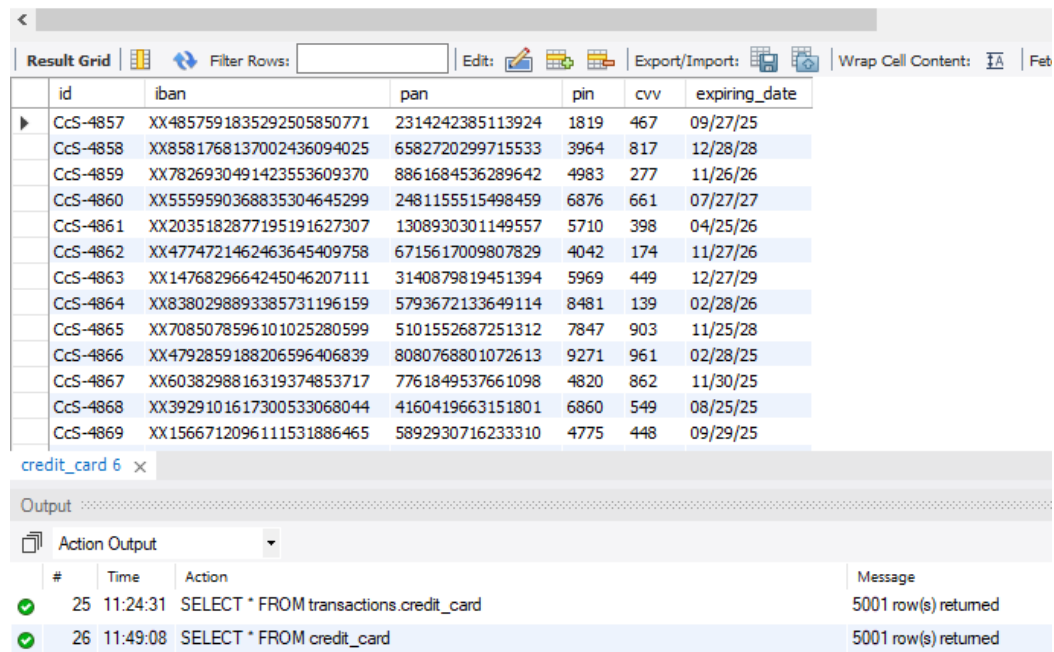
| #    | Time     | Action                                                   | Message           |
|------|----------|----------------------------------------------------------|-------------------|
| ✓ 21 | 11:15:58 | SELECT* FROM transaction WHERE id='CcU-9999'             | 0 row(s) returned |
| ✓ 22 | 11:16:33 | SELECT* FROM transaction WHERE credit_card_id='CcU-9999' | 1 row(s) returned |

## Ejercicio 4

Desde recursos humanos te solicitan eliminar la columna "pan" de la tabla credit\_card. Recuerda mostrar el cambio realizado.

- Tabla credit\_card antes del cambio:

```
120      #eliminar la columna "pan" de la tabla credit_card
121      #tabla original
122      • SELECT *
123      FROM credit_card;
124
```



The screenshot shows a database management interface. At the top, there's a toolbar with icons for filtering, editing, and exporting. Below the toolbar is a table with 7 columns: id, iban, pan, pin, cvv, and expiring\_date. The table contains 15 rows of data. Below the table, there's a tab labeled 'credit\_card 6'. Underneath the tab, there's an 'Output' section with a dropdown menu set to 'Action Output'. Below this, there's a table with 4 columns: #, Time, Action, and Message. It shows two actions: a SELECT query on 'transactions.credit\_card' at 11:24:31 and a SELECT query on 'credit\_card' at 11:49:08, both returning 5001 rows.

|   | id       | iban                     | pan              | pin  | cvv | expiring_date |
|---|----------|--------------------------|------------------|------|-----|---------------|
| ▶ | CcS-4857 | XX4857591835292505850771 | 2314242385113924 | 1819 | 467 | 09/27/25      |
|   | CcS-4858 | XX8581768137002436094025 | 6582720299715533 | 3964 | 817 | 12/28/28      |
|   | CcS-4859 | XX7826930491423553609370 | 8861684536289642 | 4983 | 277 | 11/26/26      |
|   | CcS-4860 | XX5559590368835304645299 | 2481155515498459 | 6876 | 661 | 07/27/27      |
|   | CcS-4861 | XX2035182877195191627307 | 1308930301149557 | 5710 | 398 | 04/25/26      |
|   | CcS-4862 | XX4774721462463645409758 | 6715617009807829 | 4042 | 174 | 11/27/26      |
|   | CcS-4863 | XX1476829664245046207111 | 3140879819451394 | 5969 | 449 | 12/27/29      |
|   | CcS-4864 | XX8380298893385731196159 | 5793672133649114 | 8481 | 139 | 02/28/26      |
|   | CcS-4865 | XX7085078596101025280599 | 5101552687251312 | 7847 | 903 | 11/25/28      |
|   | CcS-4866 | XX4792859188206596406839 | 8080768801072613 | 9271 | 961 | 02/28/25      |
|   | CcS-4867 | XX6038298816319374853717 | 7761849537661098 | 4820 | 862 | 11/30/25      |
|   | CcS-4868 | XX3929101617300533068044 | 4160419663151801 | 6860 | 549 | 08/25/25      |
|   | CcS-4869 | XX1566712096111531886465 | 5892930716233310 | 4775 | 448 | 09/29/25      |

| #    | Time     | Action                                 | Message              |
|------|----------|----------------------------------------|----------------------|
| ✓ 25 | 11:24:31 | SELECT * FROM transactions.credit_card | 5001 row(s) returned |
| ✓ 26 | 11:49:08 | SELECT * FROM credit_card              | 5001 row(s) returned |

- Eliminación de la columna y como quedó la tabla después del cambio:

```

126 • ALTER TABLE credit_card
127 DROP COLUMN pan
128 ;
129 #revisamos si se realizaron los cambios correctamente
130 • SELECT *
131 FROM credit_card
132 ;

```

| Result Grid        |          |                          |      |     |               |  |
|--------------------|----------|--------------------------|------|-----|---------------|--|
| Filter Rows:       |          |                          |      |     |               |  |
| Edit:              |          |                          |      |     |               |  |
| Export/Import:     |          |                          |      |     |               |  |
| Wrap Cell Content: |          |                          |      |     |               |  |
| Fetch rows:        |          |                          |      |     |               |  |
|                    | id       | iban                     | pin  | cvv | expiring_date |  |
| ▶                  | CcS-4857 | XX4857591835292505850771 | 1819 | 467 | 09/27/25      |  |
|                    | CcS-4858 | XX8581768137002436094025 | 3964 | 817 | 12/28/28      |  |
|                    | CcS-4859 | XX7826930491423553609370 | 4983 | 277 | 11/26/26      |  |
|                    | CcS-4860 | XX5559590368835304645299 | 6876 | 661 | 07/27/27      |  |
|                    | CcS-4861 | XX2035182877195191627307 | 5710 | 398 | 04/25/26      |  |
|                    | CcS-4862 | XX4774721462463645409758 | 4042 | 174 | 11/27/26      |  |
|                    | CcS-4863 | XX1476829664245046207111 | 5969 | 449 | 12/27/29      |  |
|                    | CcS-4864 | XX8380298893385731196159 | 8481 | 139 | 02/28/26      |  |
|                    | CcS-4865 | XX7085078596101025280599 | 7847 | 903 | 11/25/28      |  |

credit\_card 7 x

| Output        |          |                                         |                                                        |
|---------------|----------|-----------------------------------------|--------------------------------------------------------|
| Action Output |          |                                         |                                                        |
| #             | Time     | Action                                  | Message                                                |
| ✓ 27          | 11:51:37 | ALTER TABLE credit_card DROP COLUMN pan | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 |
| ✓ 28          | 11:52:01 | SELECT * FROM credit_card               | 5001 row(s) returned                                   |

## Nivel 2

### Ejercicio 1

Elimina de la tabla transacción el registro con ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD de la base de datos.

- Antes

```

136 #Elimina de la tabla transacción el registro con ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD de la base de datos.
137 • SELECT *
138 FROM transaction
139 WHERE id = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD'
140 ;

```

| Result Grid        |                                      |                |            |         |         |           |                     |        |          |
|--------------------|--------------------------------------|----------------|------------|---------|---------|-----------|---------------------|--------|----------|
| Filter Rows:       |                                      |                |            |         |         |           |                     |        |          |
| Edit:              |                                      |                |            |         |         |           |                     |        |          |
| Export/Import:     |                                      |                |            |         |         |           |                     |        |          |
| Wrap Cell Content: |                                      |                |            |         |         |           |                     |        |          |
|                    | id                                   | credit_card_id | company_id | user_id | lat     | longitude | timestamp           | amount | declined |
| ▶                  | 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD | CcS-5019       | b-2370     | 438     | 41.5972 | 12.2218   | 2016-12-21 20:07:18 | 155.63 | 0        |
| •                  | NULL                                 | NULL           | NULL       | NULL    | NULL    | NULL      | NULL                | NULL   | NULL     |

transaction 8 x

| Output        |          |                                                                             |                      |
|---------------|----------|-----------------------------------------------------------------------------|----------------------|
| Action Output |          |                                                                             |                      |
| #             | Time     | Action                                                                      | Message              |
| ✓ 28          | 11:52:01 | SELECT * FROM credit_card                                                   | 5001 row(s) returned |
| ✓ 29          | 11:56:49 | SELECT * FROM transaction WHERE id = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD' | 1 row(s) returned    |

- Después

```

142     #eliminamos
143     • DELETE FROM transaction
144     WHERE id = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD'
145     ;
146     #revisamos que se haya realizado el cambio correctamente
147     • SELECT *
148     FROM transaction
149     WHERE id = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD'
150     ;

```

| Result Grid                                  |      |                |            |         |      |           |           |        |          |
|----------------------------------------------|------|----------------|------------|---------|------|-----------|-----------|--------|----------|
| Filter Rows: <input type="text"/>            |      |                |            |         |      |           |           |        |          |
| Edit:    Export/Import:   Wrap Cell Content: |      |                |            |         |      |           |           |        |          |
|                                              | id   | credit_card_id | company_id | user_id | lat  | longitude | timestamp | amount | declined |
| *                                            | NULL | NULL           | NULL       | NULL    | NULL | NULL      | NULL      | NULL   | NULL     |

| transaction 9 x |          |                                                                             |                   |  |
|-----------------|----------|-----------------------------------------------------------------------------|-------------------|--|
| Output          |          |                                                                             |                   |  |
| Action Output   |          |                                                                             |                   |  |
| #               | Time     | Action                                                                      | Message           |  |
| ✓ 30            | 12:00:23 | DELETE FROM transaction WHERE id = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD'   | 1 row(s) affected |  |
| ✓ 31            | 12:01:09 | SELECT * FROM transaction WHERE id = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD' | 0 row(s) returned |  |

## Ejercicio 2

- Crear una vista que proporcione detalles clave sobre las compañías y sus transacciones. Será necesario que crees una vista llamada VistaMarketing que contenga la siguiente información:
  - Nombre de la compañía
  - Teléfono de contacto.
  - País de residencia.
  - Media de compra realizada por cada compañía.

Presenta la vista creada, ordenando los datos de mayor a menor promedio de compra.

```

170      #creat view
171 •    CREATE VIEW VistaMarketing AS
172      SELECT company_name, phone, country, ROUND(AVG(amount),2) as media_ventas
173      FROM company as c
174      JOIN transaction as t
175      ON c.id = t.company_id
176      GROUP BY company_name, phone, country
177      ORDER BY media_ventas DESC;
178      #revisamos la view creada
179 •    SELECT *
180      FROM VistaMarketing;

```

| Result Grid                                                                                                                                                                                      |                           |                |               |              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------|---------------|--------------|
| Filter Rows: <input type="text"/>                                                                                                                                                                |                           |                |               |              |
| Export:  Wrap Cell Content:  |                           |                |               |              |
|                                                                                                                                                                                                  | company_name              | phone          | country       | media_ventas |
| ▶                                                                                                                                                                                                | Ac Fermentum Incorporated | 06 85 56 52 33 | Germany       | 284.87       |
|                                                                                                                                                                                                  | Pretium Neque Corp.       | 07 77 48 55 28 | Australia     | 276.16       |
|                                                                                                                                                                                                  | Urna Convallis Associates | 06 01 24 77 04 | United States | 274.24       |
|                                                                                                                                                                                                  | At Associates             | 09 56 61 10 65 | New Zealand   | 272.21       |
|                                                                                                                                                                                                  | Metus Vitae Associates    | 08 25 44 40 66 | Australia     | 270.08       |
|                                                                                                                                                                                                  | Aliquet Diam Limited      | 02 76 61 47 46 | United States | 269.60       |

VistaMarketing 12 x

Output

Action Output

| #    | Time     | Action                                                                        | Message             |
|------|----------|-------------------------------------------------------------------------------|---------------------|
| ✓ 42 | 12:47:06 | CREATE VIEW VistaMarketing AS SELECT company_name, phone, country, ROUND(A... | 0 row(s) affected   |
| ✓ 43 | 12:47:09 | SELECT * FROM VistaMarketing                                                  | 101 row(s) returned |

### Ejercicio 3

Filtra la vista VistaMarketing para mostrar sólo las compañías que tienen su país de residencia en "Germany"

```
188 #Filtra VistaMarketing sólo las compañías en "Germany"
189 • ALTER VIEW VistaMarketing AS
190 SELECT company_name, phone, country, ROUND(AVG(amount),2) as media_ventas
191 FROM company as c
192 JOIN transaction as t
193 ON c.id = t.company_id
194 WHERE country = 'Germany'
195 GROUP BY company_name, phone, country
196 ORDER BY media_ventas DESC;
197 #revisión del filtro de la vista
198 • SELECT * FROM VistaMarketing;
```

< **Result Grid** | | Filter Rows:  | Export: | Wrap Cell Content:

|   | company_name               | phone          | country | media_ventas |
|---|----------------------------|----------------|---------|--------------|
| ▶ | Ac Fermentum Incorporated  | 06 85 56 52 33 | Germany | 284.87       |
|   | Nunc Interdum Incorporated | 05 18 15 48 13 | Germany | 259.32       |
|   | Convallis In Incorporated  | 06 66 57 29 50 | Germany | 257.75       |
|   | Ac Industries              | 09 34 65 40 60 | Germany | 255.15       |
|   | Rutrum Non Inc.            | 02 66 31 61 09 | Germany | 255.14       |
|   | Auctor Mauris Corp.        | 05 62 87 14 41 | Germany | 254.77       |

VistaMarketing 14 x

Output

Action Output

|   | #  | Time     | Action                                                                        | Message           |
|---|----|----------|-------------------------------------------------------------------------------|-------------------|
| ✓ | 44 | 13:14:48 | ALTER VIEW VistaMarketing AS SELECT company_name, phone, country, ROUND(AV... | 0 row(s) affected |
| ✓ | 45 | 13:14:52 | SELECT * FROM VistaMarketing                                                  | 8 row(s) returned |

Vale después del p2p, me corrigieron que el view es una tabla temporal por lo que en lugar de usar el ALTER VIEW podía usar el SELECT y ahorrarme varias líneas de código como se muestra a continuación:

```

163     #CORRECCION SERIA EL WHERE EN LA VISTA Y NO EN EL JOIN
164
165     • SELECT *
166     FROM VistaMarketing
167     WHERE country = 'Germany';
168

```

| company_name               | phone          | country | media_ventas |
|----------------------------|----------------|---------|--------------|
| Ac Fermentum Incorporated  | 06 85 56 52 33 | Germany | 284.87       |
| Nunc Interdum Incorporated | 05 18 15 48 13 | Germany | 259.32       |
| Convallis In Incorporated  | 06 66 57 29 50 | Germany | 257.75       |
| Ac Industries              | 09 34 65 40 60 | Germany | 255.15       |
| Rutrum Non Inc.            | 02 66 31 61 09 | Germany | 255.14       |

VistaMarketing 2 x

Output

Action Output

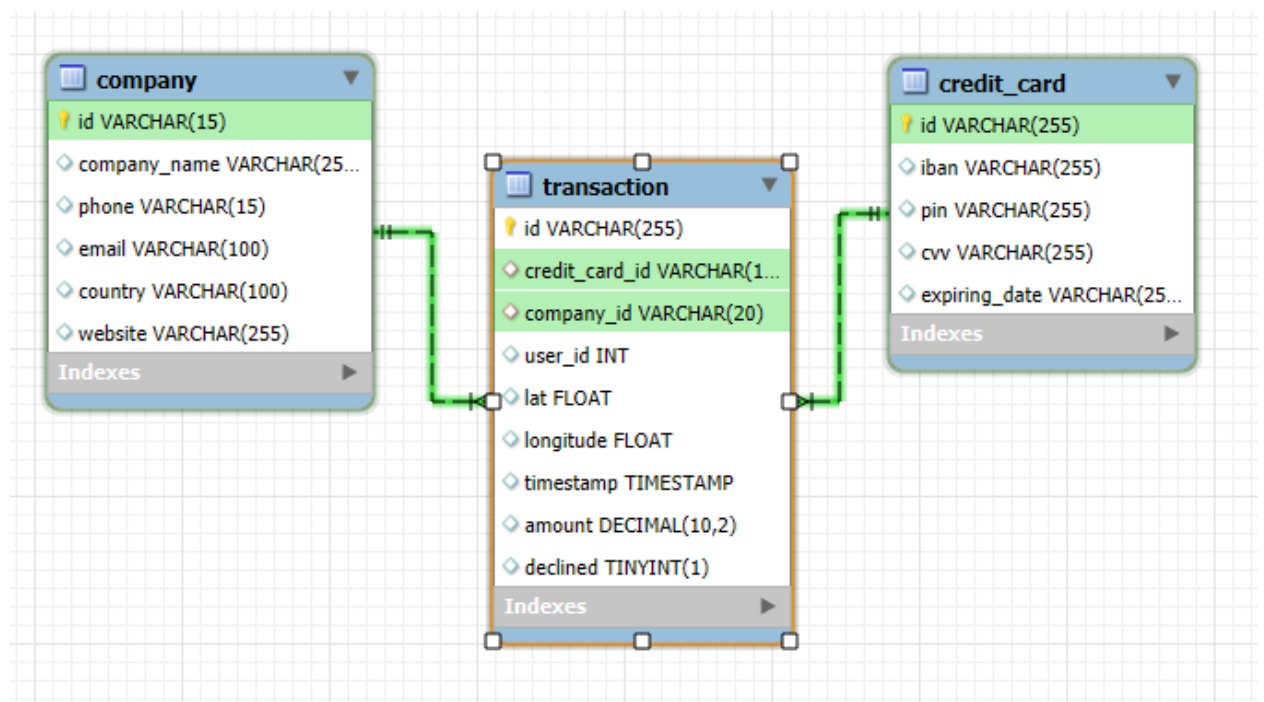
| #   | Time     | Action                                                 | Message           |
|-----|----------|--------------------------------------------------------|-------------------|
| ✓ 3 | 11:00:40 | SELECT * FROM transactions.vistamarketing              | 8 row(s) returned |
| ✓ 4 | 11:28:02 | SELECT * FROM VistaMarketing WHERE country = 'Germany' | 8 row(s) returned |

### Nivel 3

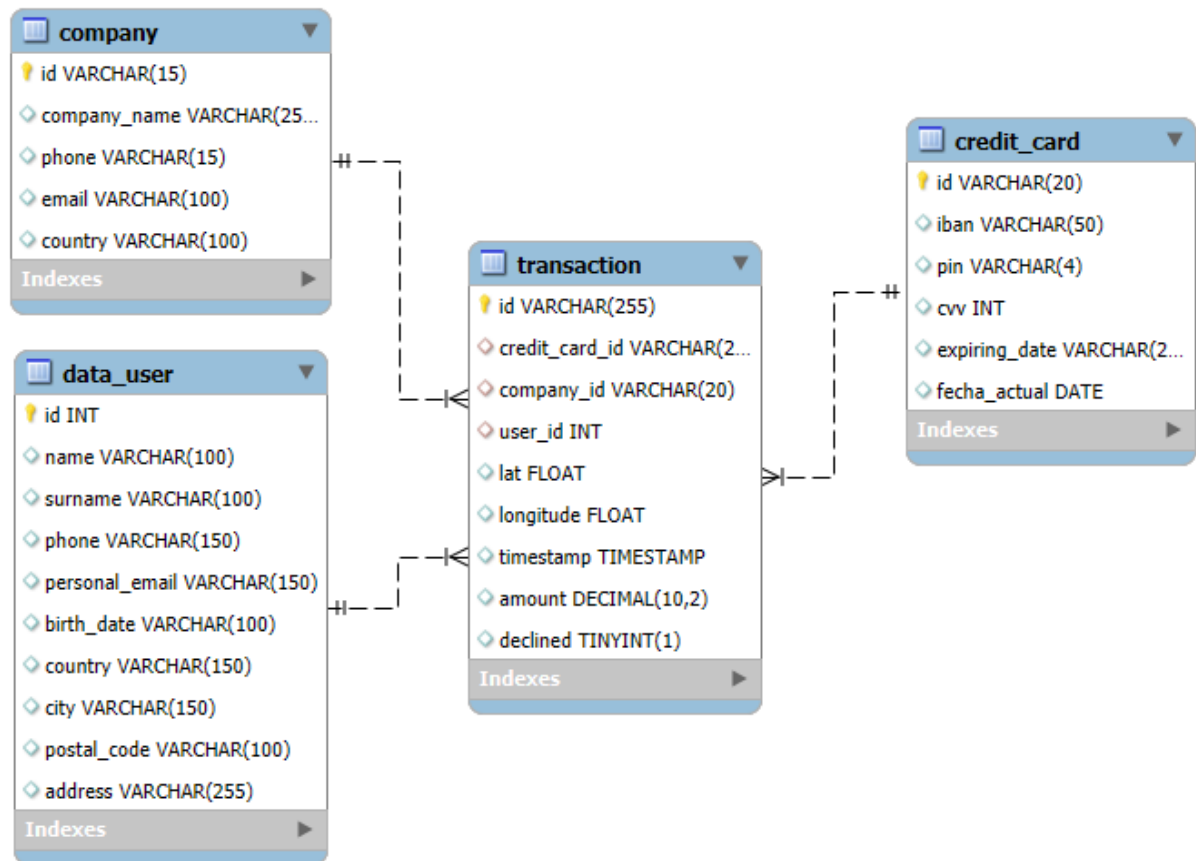
#### Ejercicio 1

El compañero me pide que le ayude con el paso a paso y los comandos ejecutados para obtener el diagrama (4 tablas).

Bueno, partiendo del modelo resultante de haber realizado todos los ejercicios anteriores, el cual se muestra a continuación:



Se generarán los cambios necesarios para llegar al modelo que muestra el compañero (mostrado a continuación) y se mostrará el paso a paso de cada uno de los cambios realizados.



- Creación de la tabla users

```
1 • CREATE TABLE IF NOT EXISTS user (  
2     id CHAR(10) PRIMARY KEY,  
3     name VARCHAR(100),  
4     surname VARCHAR(100),  
5     phone VARCHAR(150),  
6     email VARCHAR(150),  
7     birth_date VARCHAR(100),  
8     country VARCHAR(150),  
9     city VARCHAR(150),  
10    postal_code VARCHAR(100),  
11    address VARCHAR(255)  
12 );  
13
```

Output

Action Output

| #    | Time     | Action                                                                                                                                                                                                                                                       | Message           |
|------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| ✓ 46 | 13:16:13 | SELECT * FROM VistaMarketing                                                                                                                                                                                                                                 | 8 row(s) returned |
| ✓ 47 | 13:38:57 | CREATE TABLE IF NOT EXISTS user (id CHAR(10) PRIMARY KEY, name VARCHAR(100), surname VARCHAR(100), phone VARCHAR(150), email VARCHAR(150), birth_date VARCHAR(100), country VARCHAR(150), city VARCHAR(150), postal_code VARCHAR(100), address VARCHAR(255)) | 0 row(s) affected |

```
1 • SELECT * FROM transactions.user;
```

Result Grid

|   | id   | name | surname | phone | email | birth_date | country | city | postal_code | address |
|---|------|------|---------|-------|-------|------------|---------|------|-------------|---------|
| * | NULL | NULL | NULL    | NULL  | NULL  | NULL       | NULL    | NULL | NULL        | NULL    |

user 1 x

Output

Action Output

| #    | Time     | Action                                                                                                                                                                                                                                                       | Message           |
|------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| ✓ 47 | 13:38:57 | CREATE TABLE IF NOT EXISTS user (id CHAR(10) PRIMARY KEY, name VARCHAR(100), surname VARCHAR(100), phone VARCHAR(150), email VARCHAR(150), birth_date VARCHAR(100), country VARCHAR(150), city VARCHAR(150), postal_code VARCHAR(100), address VARCHAR(255)) | 0 row(s) affected |
| ✓ 48 | 13:39:33 | SELECT * FROM transactions.user                                                                                                                                                                                                                              | 0 row(s) returned |

- Cargar los datos en la tabla users

1 • `SELECT * FROM transactions.user;`

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

Fetch rows:

|   | id   | name    | surname  | phone          | email                        | birth_date   | country       | city      | postal_code | address                   |
|---|------|---------|----------|----------------|------------------------------|--------------|---------------|-----------|-------------|---------------------------|
| ▶ | 1    | Zeus    | Gamble   | 1-282-581-0551 | interdum.enim@protonmail.edu | Nov 17, 1985 | United States | New York  | 10001       | 348-7818 Sagittis St.     |
|   | 10   | Robert  | Mccarthy | (324) 746-6771 | fermentum@protonmail.com     | Apr 30, 1984 | United States | San Jose  | 95101       | P.O. Box 773              |
|   | 100  | Melodie | Mclean   | 1-677-221-7152 | risus.varius@google.ca       | Sep 15, 1989 | United States | San Jose  | 95101       | Ap #644-8492 Sagittis St. |
|   | 1000 | Amkjr   | Qbulrxbp | +48-258-9936   | amkjr.qbulrxbp@example.com   | May 17, 1970 | Germany       | Stuttgart | 70173       | 215 Qbulrxbp St           |
|   | 1001 | Nfvrlb  | Oydaiwbg | +94-121-2522   | nfvrlb.oydaiwbg@example.com  | Mar 4, 1994  | Germany       | Cologne   | 50667       | 121 Oydaiwbg St           |
|   | 1002 | Ijbfmd  | Jbddzhvp | +70-120-3668   | ijbfmd.jbddzhvp@example.com  | Sep 27, 2001 | Germany       | Munich    | 80331       | 412 Jbddzhvp St           |
|   | 1003 | Uycig   | Sfbdymzj | +58-123-6968   | uycig.sfbdymzj@example.com   | Jan 20, 1981 | Germany       | Stuttgart | 70173       | 735 Sfbdymzj St           |
|   | 1004 | Yjqrq   | Ojizvgqi | +77-944-2340   | yjqrq.ojizvgqi@example.com   | Jul 27, 1954 | Germany       | Munich    | 80331       | 685 Ojizvgqi St           |
|   | 1005 | Mnlqtu  | Glofegwk | +54-801-2627   | mnlqtu.glofegwk@example.com  | Nov 15, 1962 | Portugal      | Funchal   | 9000-001    | 8 Glofegwk St             |
|   | 1006 | Rbkocq  | Swkmynuj | +38-186-8580   | rbkocq.swkmynuj@example.com  | Dec 2, 1989  | United States | New York  | 10001       | 567 Swkmynuj St           |
|   | 1007 | Ehcrip  | Xahkzrlm | +71-862-6101   | ehcrip.xahkzrlm@example.com  | Nov 8, 1971  | Spain         | Sevilla   | 41001       | 205 Xahkzrlm St           |
|   | 1008 | Securp  | Faofvqfy | +76-822-2041   | securp.faofvqfy@example.com  | Nov 13, 1957 | Sweden        | Stockholm | 111 20      | 535 Faofvqfy St           |
|   | 1009 | Rsznah  | Vfqfqaq  | +67-294-3155   | rsznah.vfqfqaq@example.com   | Nov 10, 1952 | Spain         | Valencia  | 46001       | 507 Vfqfqaq St            |

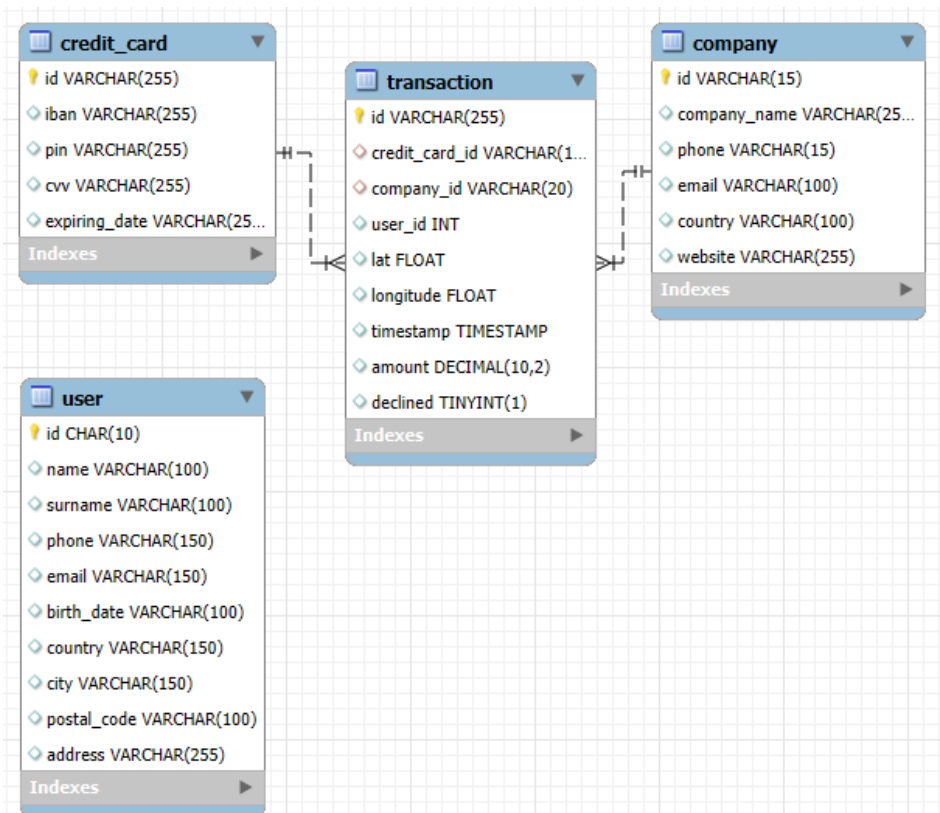
user 3

Output

Action Output

| #      | Time     | Action                                                                                      | Message              |
|--------|----------|---------------------------------------------------------------------------------------------|----------------------|
| ✓ 5048 | 13:41:28 | INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_cod... | 1 row(s) affected    |
| ✓ 5049 | 13:43:08 | SELECT * FROM transactions.user                                                             | 5000 row(s) returned |
| ✓ 5050 | 13:43:21 | SELECT * FROM transactions.user                                                             | 5000 row(s) returned |

- Se creó la tabla users y se agregaron los datos, pero no está relacionada con el resto del modelo



- Se crea la relación entre tablas para generar un modelo con las 4 tablas:

| Output        |            |                                                                              |                                                                                               |  |
|---------------|------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--|
| Action Output |            |                                                                              |                                                                                               |  |
| #             | Time       | Action                                                                       | Message                                                                                       |  |
| ✓             | 4 10:59:55 | SELECT * FROM transactions.transaction                                       | 100000 row(s) returned                                                                        |  |
| ✗             | 5 11:01:07 | ALTER TABLE transaction ADD CONSTRAINT FK_t_u FOREIGN KEY (user_id) REFER... | Error Code: 3780. Referencing column 'user_id' and referenced column 'id' in foreign key c... |  |

Lo que básicamente indica que las variables entre tablas tienen diferentes data types (<https://stackoverflow.com/questions/58550408/error-code-3780-referencing-and-referenced-column-in-foreign-key-constraint-are>). Por ende, opté por primero realizar todos los cambios de data types, nombres de variables y tablas, y agregar o eliminar las filas y columnas que sean necesarias, en cada tabla para poder realizar esta relación posteriormente.

- Tabla company:**

- antes

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

|   | id     | company_name                | phone          | email                            | country       | website                        |
|---|--------|-----------------------------|----------------|----------------------------------|---------------|--------------------------------|
| ▶ | b-2222 | Ac Fermentum Incorporated   | 06 85 56 52 33 | donec.porttitor.tellus@yahoo.net | Germany       | https://instagram.com/site     |
|   | b-2226 | Magna A Neque Industries    | 04 14 44 64 62 | risus.donec.nibh@icloud.org      | Australia     | https://whatsapp.com/group/9   |
|   | b-2230 | Fusce Corp.                 | 08 14 97 58 85 | risus@protonmail.edu             | United States | https://pinterest.com/sub/cars |
|   | b-2234 | Convallis In Incorporated   | 06 66 57 29 50 | mauris.ut@aol.co.uk              | Germany       | https://cnn.com/user/110       |
|   | b-2238 | Ante Iaculis Nec Foundation | 08 23 04 99 53 | sed.dictum.proin@outlook.ca      | New Zealand   | https://netflix.com/settings   |
|   | b-2242 | Donec Ltd                   | 01 25 51 37 37 | at.iaculis@hotmail.co.uk         | Norway        | https://nytimes.com/user/110   |

company 2

Output

Action Output

| # | Time     | Action                                                                       | Message                                       |
|---|----------|------------------------------------------------------------------------------|-----------------------------------------------|
| 8 | 11:04:53 | ALTER TABLE transaction ADD CONSTRAINT FK_t_u FOREIGN KEY (user_id) REFER... | Error Code: 3780. Referencing column 'user... |
| 9 | 11:44:43 | SELECT * FROM transactions.company                                           | 101 row(s) returned                           |

Después:

1 • `SELECT * FROM transactions.company;`

| id     | company_name                | phone          | email                            | country       |
|--------|-----------------------------|----------------|----------------------------------|---------------|
| b-2222 | Ac Fermentum Incorporated   | 06 85 56 52 33 | donec.porttitor.tellus@yahoo.net | Germany       |
| b-2226 | Magna A Neque Industries    | 04 14 44 64 62 | risus.donec.nibh@icloud.org      | Australia     |
| b-2230 | Fusce Corp.                 | 08 14 97 58 85 | risus@protonmail.edu             | United States |
| b-2234 | Convallis In Incorporated   | 06 66 57 29 50 | mauris.ut@aol.couk               | Germany       |
| b-2238 | Ante Iaculis Nec Foundation | 08 23 04 99 53 | sed.dictum.proin@outlook.ca      | New Zealand   |
| b-2242 | Donec Ltd                   | 01 25 51 37 37 | at.iaculis@hotmail.couk          | Norway        |

company 3 x

Output

| #    | Time     | Action                                  | Message                                                |
|------|----------|-----------------------------------------|--------------------------------------------------------|
| ✓ 10 | 11:45:46 | ALTER TABLE company DROP COLUMN website | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 |
| ✓ 11 | 11:46:03 | SELECT * FROM transactions.company      | 101 row(s) returned                                    |

Los tipos de datos de esta tabla coinciden con los presentados en el modelo del compañero, así que este era el único cambio por realizar en la tabla company.

### Continuamos con los cambios a realizar en la tabla transaction:

La única diferencia que veo por cambiar es el tipo de datos del ID de tarjeta de crédito, en la tabla transacción, pues el modelo del compañero pareciera tenerlo en VARCHAR 255 y en mi modelo está VARCHAR 15, en verdad no sé si sea un cambio completamente necesario de realizar, pero es una operación muy sencilla. En realidad, si nos fijamos un poco más allá, este campo es el fk en la tabla company que en el modelo del compañero está en VARCHAR 20, así que pondremos en ambas columnas en ambas tablas con ese tipo de dato.

```

210 #cambio de data type tabla transaction
211 # credit_card_id de VARCHAR 255 a 20
212 • ALTER TABLE transaction
213     MODIFY credit_card_id VARCHAR (20);
214
215
216
217 #union de la tabla user al modelo ALTER TABLE transaction
218 • ALTER TABLE transaction

```

Output

| #    | Time     | Action                                                           | Message                                                                                    |
|------|----------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| ✗ 12 | 12:02:04 | ALTER TABLE nombre_tabla ALTER COLUMN nombre_columna VARCHAR(20) | Error Code: 1064. You have an error in your SQL syntax; check the manual that correspon... |
| ✓ 13 | 12:04:00 | ALTER TABLE transaction MODIFY credit_card_id VARCHAR (20)       | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0                                     |

Para corroborar de que los cambios se realizaron correctamente:

Sprint 3 MMCG\*

company

credit\_card

user

transaction

estructura datos

transactions.transaction

Info

Columns

Indexes

Triggers

Foreign keys

Partitions

Grants

DDL

| Column         | Type          | Default Value | Nullable | Character Set | Collation        | Privileges                      | Extra | Comments |
|----------------|---------------|---------------|----------|---------------|------------------|---------------------------------|-------|----------|
| amount         | decimal(10,2) |               | YES      |               |                  | select,insert,update,references |       |          |
| company_id     | varchar(20)   |               | YES      | utf8mb4       | utf8mb4_0900_... | select,insert,update,references |       |          |
| credit_card_id | varchar(20)   |               | YES      | utf8mb4       | utf8mb4_0900_... | select,insert,update,references |       |          |
| declined       | tinyint(1)    |               | YES      |               |                  | select,insert,update,references |       |          |
| id             | varchar(255)  |               | NO       | utf8mb4       | utf8mb4_0900_... | select,insert,update,references |       |          |
| lat            | float         |               | YES      |               |                  | select,insert,update,references |       |          |
| longitude      | float         |               | YES      |               |                  | select,insert,update,references |       |          |
| timestamp      | timestamp     |               | YES      |               |                  | select,insert,update,references |       |          |
| user_id        | int           |               | YES      |               |                  | select,insert,update,references |       |          |

Count: 9

Output

Action Output

| #  | Time     | Action                                                           | Message                                                                                    |
|----|----------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 12 | 12:02:04 | ALTER TABLE nombre_tabla ALTER COLUMN nombre_columna VARCHAR(20) | Error Code: 1064. You have an error in your SQL syntax; check the manual that correspon... |
| 13 | 12:04:00 | ALTER TABLE transaction MODIFY credit_card_id VARCHAR (20)       | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0                                     |

Vale, pues ese era el único cambio por realizar en la tabla transacción, de momento, por lo que voy a continuar con el cambio del tipo de dato del id de la tabla credit card, que es el fk de la columna anterior que se acaba de modificar en la tabla transaction, entonces le pondré el mismo tipo de variable.

```

167 #cambio de data type tabla transaction
168 # credit_card_id de VARCHAR 255 a 20
169 • ALTER TABLE transaction
170 MODIFY credit_card_id VARCHAR (20);
171

```

| Output        |          |                                                            |                                                        |
|---------------|----------|------------------------------------------------------------|--------------------------------------------------------|
| Action Output |          |                                                            |                                                        |
| #             | Time     | Action                                                     | Message                                                |
| 1             | 10:21:40 | ALTER TABLE transaction MODIFY credit_card_id VARCHAR (20) | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 |

Vale, aproveché el mismo query anterior para realizar los cambios de tipos de datos que fueran necesarios para esta tabla, así:

```

215 #cambios de data type tabla company
216 • ALTER TABLE company
217 MODIFY id VARCHAR (20),
218 MODIFY iban VARCHAR (50),
219 MODIFY pin VARCHAR (4),
220 MODIFY cvv INT
221 ;
222
223
224

```

| Output        |          |                                                            |                                                        |
|---------------|----------|------------------------------------------------------------|--------------------------------------------------------|
| Action Output |          |                                                            |                                                        |
| #             | Time     | Action                                                     | Message                                                |
| 13            | 12:04:00 | ALTER TABLE transaction MODIFY credit_card_id VARCHAR (20) | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 |
| 14            | 12:19:28 | ALTER TABLE company MODIFY id VARCHAR (20)                 | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 |

Vale pues aquí me empecé a encontrar con que no se podían realizar estos cambios debido a las relaciones que hay entre las tablas, en este caso las modificaciones que se van a realizar son simples por lo que se optó por eliminar las fk constraint entre las tablas, realizar los cambios necesarios y luego volver a armar el modelo. a continuación, se muestran los screen shots y el paso a paso que se siguió para hacerlo:

## 1. Elimine las fk

```

---
217 #Vale en el camino me encontré con varias liadasa entonces para
218 #evitar estos lios, se eliminan las forein keys, se realizan todos los cambios necesarios
219 #y se vuelve a restablecer las relaciones para completar el modelo
220
221 #eliminacion de fk
222 • ALTER TABLE transaction
223 DROP CONSTRAINT FK_t_cc
224 ;
225 • ALTER TABLE transaction
226 DROP CONSTRAINT transaction_ibfk_1
227 ;
228
229 #cambios de data type tabla company
230 • ALTER TABLE credit_card
231 MODIFY id VARCHAR (20),
232 MODIFY iban VARCHAR (50),
233 MODIFY pin VARCHAR (4),
234 MODIFY cvv INT
235 ;

```

Output

Action Output

| #    | Time     | Action                                                     | Message                                                |
|------|----------|------------------------------------------------------------|--------------------------------------------------------|
| ✓ 26 | 13:58:34 | ALTER TABLE transaction DROP CONSTRAINT FK_t_cc            | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 |
| ✓ 27 | 13:59:25 | ALTER TABLE transaction DROP CONSTRAINT transaction_ibfk_1 | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 |

## 2. Realizar los cambios necesarios

```

221 #eliminacion de fk
222 • ALTER TABLE transaction
223 DROP CONSTRAINT FK_t_cc
224 ;
225 • ALTER TABLE transaction
226 DROP CONSTRAINT transaction_ibfk_1
227 ;
228
229 #cambios de data type tabla company
230 • ALTER TABLE credit_card
231 MODIFY id VARCHAR (20),
232 MODIFY iban VARCHAR (50),
233 MODIFY pin VARCHAR (4),
234 MODIFY cvv INT
235 ;
236
237
238 #union de la tabla user al modelo ALTER TABLE transaction

```

Output

Action Output

| #    | Time     | Action                                                                         | Message                                                      |
|------|----------|--------------------------------------------------------------------------------|--------------------------------------------------------------|
| ✓ 27 | 13:59:25 | ALTER TABLE transaction DROP CONSTRAINT transaction_ibfk_1                     | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0       |
| ✓ 28 | 14:01:08 | ALTER TABLE credit_card MODIFY id VARCHAR (20), MODIFY iban VARCHAR (50), M... | 5001 row(s) affected Records: 5001 Duplicates: 0 Warnings: 0 |

Comprobamos:

|                                                                                                     |              |               |          |               |                  |                                 |       |          |
|-----------------------------------------------------------------------------------------------------|--------------|---------------|----------|---------------|------------------|---------------------------------|-------|----------|
| Sprint 3 MMCG                                                                                       |              |               |          |               |                  |                                 |       |          |
| company credit_card user transaction estructura dades transactions.company transactions.credit_card |              |               |          |               |                  |                                 |       |          |
| Info Columns Indexes Triggers Foreign keys Partitions Grants DDL                                    |              |               |          |               |                  |                                 |       |          |
| Column                                                                                              | Type         | Default Value | Nullable | Character Set | Collation        | Privileges                      | Extra | Comments |
| cvv                                                                                                 | int          |               | YES      | utf8mb4       | utf8mb4_0900_... | select,insert,update,references |       |          |
| expiring_date                                                                                       | varchar(255) |               | YES      | utf8mb4       | utf8mb4_0900_... | select,insert,update,references |       |          |
| iban                                                                                                | varchar(50)  |               | YES      | utf8mb4       | utf8mb4_0900_... | select,insert,update,references |       |          |
| id                                                                                                  | varchar(20)  |               | NO       | utf8mb4       | utf8mb4_0900_... | select,insert,update,references |       |          |
| pin                                                                                                 | varchar(4)   |               | YES      | utf8mb4       | utf8mb4_0900_... | select,insert,update,references |       |          |

Count: 5

Output

| #  | Time     | Action                                                                         | Message                                                      | Duration / Fetch |
|----|----------|--------------------------------------------------------------------------------|--------------------------------------------------------------|------------------|
| 27 | 13:59:25 | ALTER TABLE transaction DROP CONSTRAINT transaction_ibfk_1                     | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0       | 0.031 sec        |
| 28 | 14:01:08 | ALTER TABLE credit_card MODIFY id VARCHAR (20), MODIFY iban VARCHAR (50), M... | 5001 row(s) affected Records: 5001 Duplicates: 0 Warnings: 0 | 0.359 sec        |

Añadimos la columna que falta: fecha actual aprovechando el mismo query

|     |                                                           |
|-----|-----------------------------------------------------------|
| 223 | DROP CONSTRAINT FK_t_cc                                   |
| 224 | ;                                                         |
| 225 | ALTER TABLE transaction                                   |
| 226 | DROP CONSTRAINT transaction_ibfk_1                        |
| 227 | ;                                                         |
| 228 |                                                           |
| 229 | #cambios de data type tabla company                       |
| 230 | ALTER TABLE credit_card                                   |
| 231 | ADD COLUMN fecha_actual DATE,                             |
| 232 | MODIFY id VARCHAR (20),                                   |
| 233 | MODIFY iban VARCHAR (50),                                 |
| 234 | MODIFY pin VARCHAR (4),                                   |
| 235 | MODIFY cvv INT                                            |
| 236 | ;                                                         |
| 237 |                                                           |
| 238 |                                                           |
| 239 | #union de la tabla user al modelo ALTER TABLE transaction |
| 240 | ALTER TABLE transaction                                   |
| 241 | ADD CONSTRAINT FK_t_u                                     |
| 242 | FOREIGN KEY (user_id)                                     |

Output

| #  | Time     | Action                                 | Message              | Duration / Fetch      |
|----|----------|----------------------------------------|----------------------|-----------------------|
| 41 | 14:13:34 | SELECT * FROM transactions.credit_card | 5001 row(s) returned | 0.016 sec / 0.016 sec |
| 42 | 14:14:33 | SELECT * FROM transactions.credit_card | 5001 row(s) returned | 0.000 sec / 0.031 sec |

Vale ahora van todas las modificaciones de la tabla users

- Cambiar el nombre de la tabla de user a data\_user:

```

227
228 #cambios de data type tabla company
229 • ALTER TABLE credit_card
230 ADD COLUMN fecha_actual DATE,
231 MODIFY id VARCHAR (20),
232 MODIFY iban VARCHAR (50),
233 MODIFY pin VARCHAR (4),
234 MODIFY cvv INT
235 ;
236
237 #modificaciones de la tabla user
238 • ALTER TABLE user
239 RENAME TO data_user,
240 MODIFY id INT
241 ;
242
243

```

| Output        |          |                                                     |                                                              |
|---------------|----------|-----------------------------------------------------|--------------------------------------------------------------|
| Action Output |          |                                                     |                                                              |
| #             | Time     | Action                                              | Message                                                      |
| ✓ 44          | 14:28:30 | ALTER TABLE user RENAME TO data_user, MODIFY id INT | 5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0 |
| ✗ 45          | 14:28:45 | SELECT * FROM transactions.user                     | Error Code: 1146. Table 'transactions.user' doesn't exist    |
| ✓ 46          | 14:28:55 | SELECT * FROM transactions.data_user                | 5000 row(s) returned                                         |

- Id pasarlo a int y cambiar el nombre de la columna de email a personal\_email

```

228 #cambios de data type tabla company
229 • ALTER TABLE credit_card
230 ADD COLUMN fecha_actual DATE,
231 MODIFY id VARCHAR (20),
232 MODIFY iban VARCHAR (50),
233 MODIFY pin VARCHAR (4),
234 MODIFY cvv INT
235 ;
236
237 #modificaciones de la tabla user
238 • ALTER TABLE data_user
239 RENAME TO data_user,
240 MODIFY id INT,
241 RENAME COLUMN email TO personal_email
242 ;
243

```

| Output        |          |                                                                             |                                                           |
|---------------|----------|-----------------------------------------------------------------------------|-----------------------------------------------------------|
| Action Output |          |                                                                             |                                                           |
| #             | Time     | Action                                                                      | Message                                                   |
| ✓ 46          | 14:28:55 | SELECT * FROM transactions.data_user                                        | 5000 row(s) returned                                      |
| ✗ 47          | 14:31:44 | ALTER TABLE user RENAME TO data_user, MODIFY id INT, RENAME COLUMN email... | Error Code: 1146. Table 'transactions.user' doesn't exist |
| ✓ 48          | 14:32:00 | ALTER TABLE data_user RENAME TO data_user, MODIFY id INT, RENAME COLUMN ... | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0    |

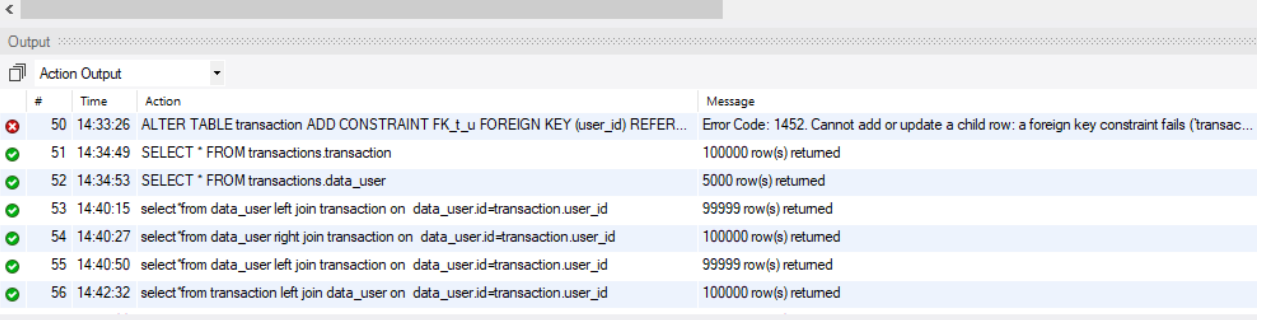
- Volvemos a establecer las relaciones con las fk desde la tabla transaction hacia las demás tablas:

Al establecer la relación transaction-data\_user arroja un error que básicamente dice que en la tabla transaction hay un registro que no existe en la tabla data\_user. Para corregir este error, primero hice una left join para identificar el registro faltante, y añadir una fila nueva a la tabla data\_user como se muestra en los dos siguientes screenshots:

```

245 #union de la tabla user al modelo ALTER TABLE transaction
246 • ALTER TABLE transaction
247 ADD CONSTRAINT FK_t_u
248 FOREIGN KEY (user_id)
249 REFERENCES data_user(id);
250
251 #me arroja un error diciendo que hay un registro en transaction que no existe en data_user
252 #como corroborarlo?
253 #left join?
254 • select*from transaction
255 left join data_user on
256 data_user.id=transaction.user_id
257 where data_user.id is NULL

```



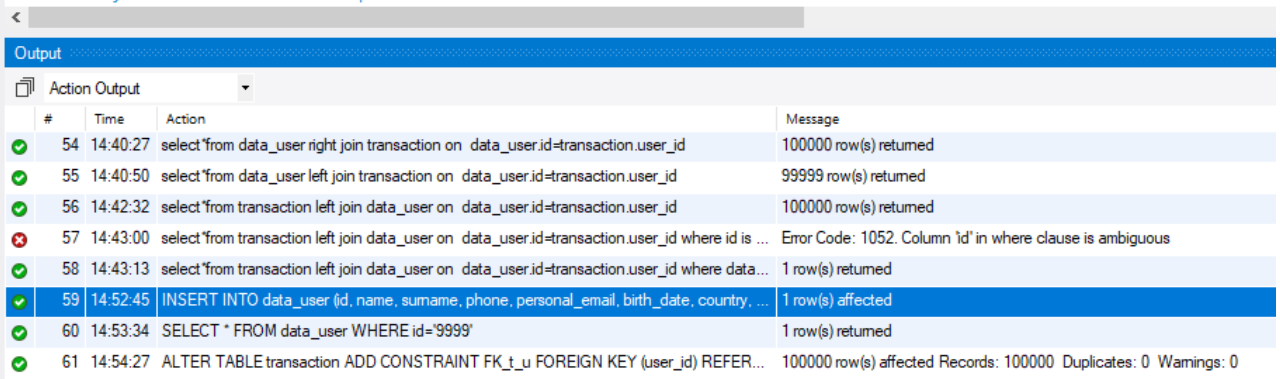
| #  | Time     | Action                                                                           | Message                                                                                        |
|----|----------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| 50 | 14:33:26 | ALTER TABLE transaction ADD CONSTRAINT FK_t_u FOREIGN KEY (user_id) REFER...     | Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails (transac... |
| 51 | 14:34:49 | SELECT * FROM transactions.transaction                                           | 100000 row(s) returned                                                                         |
| 52 | 14:34:53 | SELECT * FROM transactions.data_user                                             | 5000 row(s) returned                                                                           |
| 53 | 14:40:15 | select*from data_user left join transaction on data_user.id=transaction.user_id  | 99999 row(s) returned                                                                          |
| 54 | 14:40:27 | select*from data_user right join transaction on data_user.id=transaction.user_id | 100000 row(s) returned                                                                         |
| 55 | 14:40:50 | select*from data_user left join transaction on data_user.id=transaction.user_id  | 99999 row(s) returned                                                                          |
| 56 | 14:42:32 | select*from transaction left join data_user on data_user.id=transaction.user_id  | 100000 row(s) returned                                                                         |

- añadir la nueva fila y comprobación de que se realizó la acción correctamente:

```

260 #claro fue la info que se añadió en niveles anteriores, por ende
261 #se añade esta fila en la tabla data_user
262
263 • INSERT INTO data_user (id, name, surname, phone, personal_email, birth_date, country, city, postal_code, address)
264 VALUES ('9999', NULL, NULL, NULL, NULL, NULL, NULL, NULL, NULL, NULL)
265 ;
266 #comprobamos:
267 • SELECT *
268 FROM data_user
269 WHERE id='9999';
270
271 #y ahora si se debería de poder unir la tabla user al modelo

```



| #  | Time     | Action                                                                                          | Message                                                          |
|----|----------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 54 | 14:40:27 | select*from data_user right join transaction on data_user.id=transaction.user_id                | 100000 row(s) returned                                           |
| 55 | 14:40:50 | select*from data_user left join transaction on data_user.id=transaction.user_id                 | 99999 row(s) returned                                            |
| 56 | 14:42:32 | select*from transaction left join data_user on data_user.id=transaction.user_id                 | 100000 row(s) returned                                           |
| 57 | 14:43:00 | select*from transaction left join data_user on data_user.id=transaction.user_id where id is ... | Error Code: 1052. Column 'id' in where clause is ambiguous       |
| 58 | 14:43:13 | select*from transaction left join data_user on data_user.id=transaction.user_id where data...   | 1 row(s) returned                                                |
| 59 | 14:52:45 | INSERT INTO data_user (id, name, sumame, phone, personal_email, birth_date, country, ...        | 1 row(s) affected                                                |
| 60 | 14:53:34 | SELECT * FROM data_user WHERE id='9999'                                                         | 1 row(s) returned                                                |
| 61 | 14:54:27 | ALTER TABLE transaction ADD CONSTRAINT FK_t_u FOREIGN KEY (user_id) REFER...                    | 100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0 |

```

222 #comprobamos:
223 • SELECT *
224 FROM data_user
225 WHERE id='9999';
226
227 #y ahora si se deberia de poder unir la tabla user al modelo
228 • ALTER TABLE transaction

```

Result Grid

|   | id   | name | surname | phone | personal_email | birth_date | country | city | postal_code | address |
|---|------|------|---------|-------|----------------|------------|---------|------|-------------|---------|
| ▶ | 9999 | NULL | NULL    | NULL  | NULL           | NULL       | NULL    | NULL | NULL        | NULL    |
| * | NULL | NULL | NULL    | NULL  | NULL           | NULL       | NULL    | NULL | NULL        | NULL    |

data\_user 1 x

Output

Action Output

| #   | Time     | Action                                                     | Message                                                |
|-----|----------|------------------------------------------------------------|--------------------------------------------------------|
| ✓ 1 | 10:21:40 | ALTER TABLE transaction MODIFY credit_card_id VARCHAR (20) | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 |
| ✓ 2 | 10:25:33 | SELECT * FROM data_user WHERE id='9999'                    | 1 row(s) returned                                      |

Ahora restablecemos todas las relaciones entre las tablas para obtener el modelo final:

```

271 #y ahora si se deberia de poder unir la tabla user al modelo
272 • ALTER TABLE transaction
273 ADD CONSTRAINT FK_t_u
274 FOREIGN KEY (user_id)
275 REFERENCES data_user(id);

```

Output

Action Output

| #    | Time     | Action                                                                                            | Message                                                          |
|------|----------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| ✓ 54 | 14:40:27 | select * from data_user right join transaction on data_user.id=transaction.user_id                | 100000 row(s) returned                                           |
| ✓ 55 | 14:40:50 | select * from data_user left join transaction on data_user.id=transaction.user_id                 | 99999 row(s) returned                                            |
| ✓ 56 | 14:42:32 | select * from transaction left join data_user on data_user.id=transaction.user_id                 | 100000 row(s) returned                                           |
| ✗ 57 | 14:43:00 | select * from transaction left join data_user on data_user.id=transaction.user_id where id is ... | Error Code: 1052. Column 'id' in where clause is ambiguous       |
| ✓ 58 | 14:43:13 | select * from transaction left join data_user on data_user.id=transaction.user_id where data...   | 1 row(s) returned                                                |
| ✓ 59 | 14:52:45 | INSERT INTO data_user (id, name, surname, phone, personal_email, birth_date, country, ...         | 1 row(s) affected                                                |
| ✓ 60 | 14:53:34 | SELECT * FROM data_user WHERE id='9999'                                                           | 1 row(s) returned                                                |
| ✓ 61 | 14:54:27 | ALTER TABLE transaction ADD CONSTRAINT FK_t_u FOREIGN KEY (user_id) REFER...                      | 100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0 |

```

271 #y ahora si se deberia de poder unir la tabla user al modelo
272 • ALTER TABLE transaction
273 ADD CONSTRAINT FK_t_u
274 FOREIGN KEY (user_id)
275 REFERENCES data_user(id);
276
277 #por ultimo volvemos a crear las fk que habiamos quitado con anterioridad para la union completa del modelo
278 #trasaction-company
279 • ALTER TABLE transaction
280 ADD CONSTRAINT FK_t_c
281 FOREIGN KEY (company_id)
282 REFERENCES company(id);
283
284 #transaction-credit_card
285 • ALTER TABLE transaction
286 ADD CONSTRAINT FK_t_cc
287 FOREIGN KEY (credit_card_id)
288 REFERENCES credit_card(id);

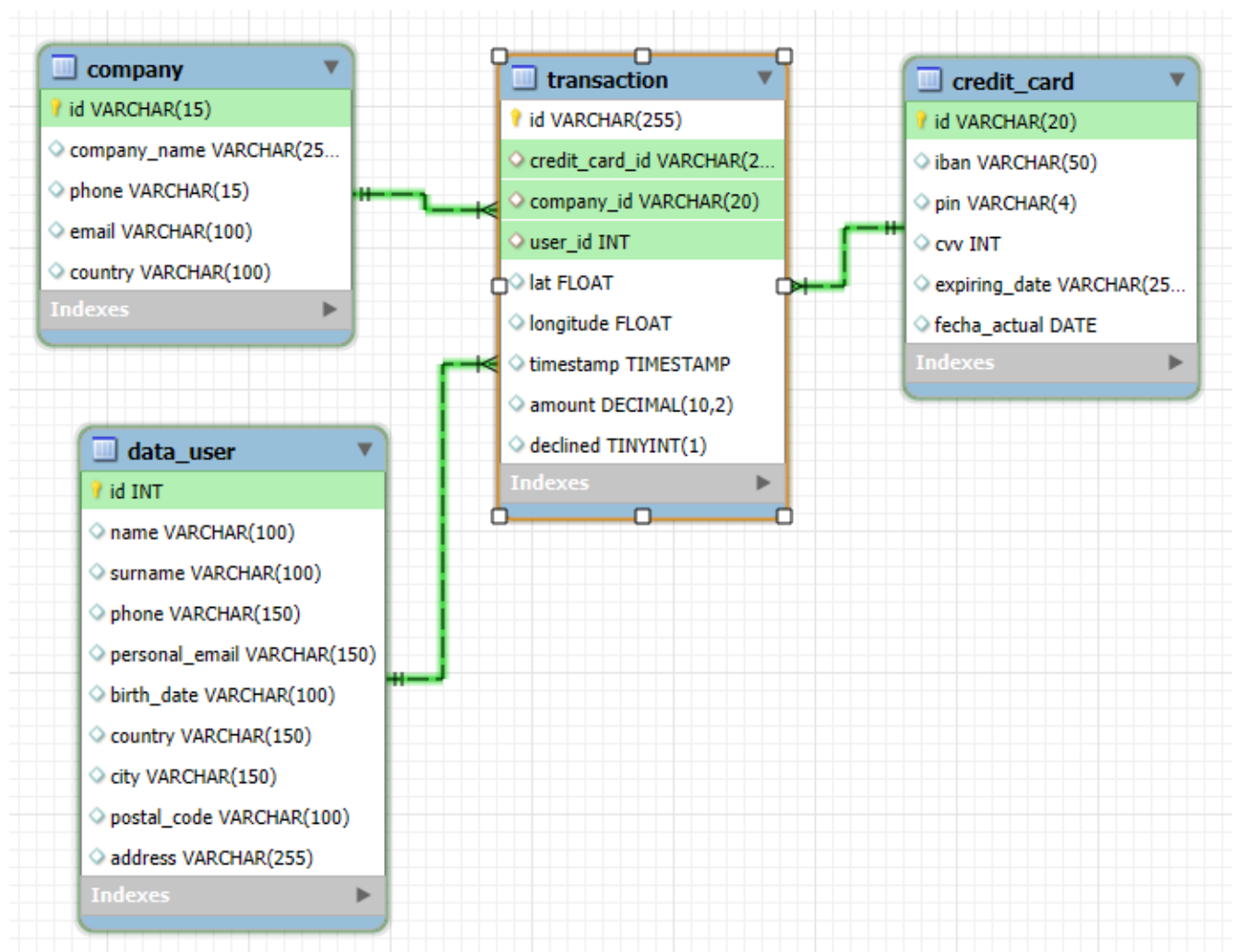
```

Output

Action Output

| #               | Time     | Action                                                                          | Message                                                          |
|-----------------|----------|---------------------------------------------------------------------------------|------------------------------------------------------------------|
| <div>✓</div> 62 | 14:59:10 | ALTER TABLE transaction ADD CONSTRAINT FK_t_c FOREIGN KEY (company_id) RE...    | 100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0 |
| <div>✓</div> 63 | 15:00:06 | ALTER TABLE transaction ADD CONSTRAINT FK_t_cc FOREIGN KEY (credit_card_id) ... | 100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0 |

Modelo resultante:



## Ejercicio 2:

La empresa también le pide crear una vista llamada "InformeTecnico" que contenga la siguiente información:

- ID de la transacción
- Nombre del usuario/a
- Apellido del usuario/a
- IBAN de la tarjeta de crédito usada.
- Nombre de la compañía de la transacción realizada.

Asegúrese de incluir información relevante de las tablas que conocerá y utilice alias para cambiar de nombre columnas según sea necesario.

Muestra los resultados de la vista, ordena los resultados de forma descendente en función de la variable ID de transacción.

- Creacion de la view:

```
258 #N3
259 #Ejercicio 2:
260 #crear vista "InformeTecnico"
261 • CREATE VIEW InformeTecnico AS
262 SELECT t.id, name, surname, iban, company_name
263 FROM transaction AS t
264 JOIN data_user AS u
265 ON t.user_id = u.id
266 JOIN credit_card AS cc
267 ON t.credit_card_id=cc.id
268 JOIN company AS c
269 ON t.company_id=c.id
270 WHERE declined =0
271 ORDER BY t.id;
272
273
```

| Output        |          |                                                                                  |                       |
|---------------|----------|----------------------------------------------------------------------------------|-----------------------|
| Action Output |          |                                                                                  |                       |
| #             | Time     | Action                                                                           | Message               |
| 9             | 11:44:32 | CREATE VIEW InformeTecnico AS SELECT t.id, name, surname, iban, company_name ... | 0 row(s) affected     |
| 10            | 11:44:46 | SELECT * FROM transactions.infometecnico                                         | 99763 row(s) returned |

- Resultado:

```
1 • SELECT * FROM transactions.infometecnico;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

| id                                   | name   | surname  | iban                       | company_name                     |
|--------------------------------------|--------|----------|----------------------------|----------------------------------|
| 00043A49-2949-494B-A5DD-A5BAE38B19DD | Gxnmjn | Fjycossj | XX691799023254713153661899 | Eget Tincidunt Dui Institute     |
| 00045D6B-ED2E-4F2F-8186-CEE074D875D0 | Wolake | Ukynumly | XX715219809262791935275891 | Neque Tellus Imperdiet Corp.     |
| 000481C3-1C26-4FEF-83A0-4CD0EB004BBD | Umhwoi | Ubmdxvmz | XX785022829713140880782522 | Fusce Corp.                      |
| 00051AA4-9CBE-4268-B070-C38062A1B3E2 | Qpfyfa | Mtnpfdfq | XX634626564536875934323485 | Mus Aenean Eget Foundation       |
| 0008A312-EDFE-4A4F-BC99-E9C92EC3CA4D | Keegan | Watson   | RO76DAFO6583348580208155   | Aliquam Iaculis Lacus Corp.      |
| 0009A151-9BCF-4E31-9053-A468FF77FAAB | Rreqrc | Jfdvcgne | XX986535778187638923658336 | Lorem Ipsum Dolor Corp.          |
| 0009D494-6245-4DF9-955D-2C084191CFFB | Novmmv | Boznhdmn | XX738074164311515717218547 | Amet Luctus Vulputate Foundation |
| 000A1DEC-CDB6-4AB2-A619-71DAB8D4A262 | Hxzzri | Mzchxymz | XX714741860105067529287700 | Nulla Integer Vulputate Corp.    |
| 000A1E64-1414-40B0-9D92-5678A4D958E2 | Ghsngl | Gkwhelrz | XX343833941783938653185254 | Auctor Mauris Corp.              |
| 000A5B7B-3473-41D8-A5C0-42D3503B543C | Luxu   | Requid   | XX3346331260637337         | Quisquam Mauris Institute        |

infometecnico 1 x

Output

Action Output

| #    | Time     | Action                                                                         | Message               |
|------|----------|--------------------------------------------------------------------------------|-----------------------|
| ✓ 9  | 11:44:32 | CREATE VIEW InfomeTecnico AS SELECT t.id, name, sumame, iban, company_name ... | 0 row(s) affected     |
| ✓ 10 | 11:44:46 | SELECT * FROM transactions.infometecnico                                       | 99763 row(s) returned |